

# Employability Readiness — QR Attendance: Setup Guide

A free, self-hosted attendance system. It runs entirely on Google (Sheets + Apps Script), writes straight to your spreadsheet, and needs no server, no hosting bill, and no app install for students — they just scan a QR with the phone camera.

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## What you get in the spreadsheet

Each check-in adds one row:

|           |                |     |        |            |              |           |           |              |        |
|-----------|----------------|-----|--------|------------|--------------|-----------|-----------|--------------|--------|
| Timestamp | Session Number | Day | Period | Student ID | Student Name | Professor | Device ID | Distance (m) | Status |
|-----------|----------------|-----|--------|------------|--------------|-----------|-----------|--------------|--------|

- **Student Name** is filled automatically from a Roster sheet (students never type it).
  - **Status** flags `ID NOT IN ROSTER` if a typed ID isn't recognised, so you can spot bad entries.
  - **Professor** is whatever the professor selected on the projection screen — students can't choose or change it.
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## One-time setup (about 15 minutes)

1. **Create the spreadsheet.** In Google Drive: New → Google Sheets. Name it e.g. *Employability Attendance*.
2. **Open the script editor.** In the sheet: **Extensions** → **Apps Script**.
3. **Add the three files.**
  - The default `Code.gs` is already open — delete its contents and paste in **Code.gs** from this package.
  - Click the + next to *Files* → **HTML** → name it exactly `Display` → paste **Display.html**.
  - + → **HTML** → name it exactly `Checkin` → paste **Checkin.html**.
  - Save (the disk icon).

#### 4. Edit the **CONFIG** block at the top of `Code.gs` :

- `SHARED_SECRET` : replace with a long random string (40+ characters). Keep it private.
- `PROFESSORS` : list the exact names you want in the pull-down.
- Set `DAYS` and `PERIODS` if different from  $10 \times 3$ .
- (Optional) For the geofence, set `REQUIRE_GEO = true` , then set `CLASS_LAT` / `CLASS_LNG` to your classroom coordinates (get them from Google Maps — right-click the building → the lat,lng at the top) and a sensible `GEO_RADIUS_M` .

#### 5. Run **setupSheets** once. In the editor, choose the function `setupSheets` from the dropdown and click **Run**. Approve the permissions prompt (it's your own account). This creates the **Attendance** and **Roster** sheets.

#### 6. **Paste your roster.** Open the **Roster** sheet → put Student IDs in column A and names in column B, starting row 2. (Remove the placeholder row.)

#### 7. **Deploy as a Web App.**

- **Deploy → New deployment → Type: Web app.**
- Execute as: **Me**.
- Who has access: **Anyone** (this is required so students can open the link without logging in). The link is unguessable; access is still gated by the rotating signed token.
- Click **Deploy**, authorise, and **copy the Web App URL**.

#### 8. **Bookmark the URL** on the classroom computer. That URL opens the professor's projection page.

If you later edit the code, use **Deploy → Manage deployments → Edit (pencil) → Version: New version → Deploy** so the live URL updates.

## Running a class

#### 1. On the classroom computer, open the Web App URL and **project it**.

2. Select your **name**, the **Day**, and the **Class/Period** → click **Show QR for this class**.
3. A large QR appears and **auto-refreshes**. Students scan it with their phone camera, enter their Student ID, and tap *Mark me present*.
4. The live counter shows arrivals in real time. When done, click **End / change class** (or reload) to switch to the next session.

That's it — rows appear in the **Attendance** sheet instantly.

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## How it resists proxy attendance

- **Rotating signed token.** The QR encodes a token that is cryptographically signed and tied to that session. It is valid only ~1–2 minutes, so a screenshot taken at the start of class is useless minutes later, and a code from one session can never be reused in another.
- **One phone, one student, per session.** A single device can't mark several people present in the same session — this stops the common "I'll sign in for my friends" move from one phone.
- **One ID per session.** Duplicate check-ins for the same ID are rejected.
- **Roster names.** Students can't invent names; the name comes from your list.
- **Optional geofence.** With `REQUIRE_GEO` on, a check-in from outside the room is rejected — this is the strongest lever against an absent friend scanning a relayed code, though indoor GPS is imperfect (see Limitations).

## Honest limitations

No scan-based system is bulletproof. A student in the room can still photograph the live QR and send it to an absent friend who scans it within the ~1–2 minute window. To do that they must coordinate in real time, every session — which is usually more effort than it's worth for a summer course. If you need to fully close that gap, turn on the geofence, and/or have a TA spot-check the room against the live counter a couple of times per class.

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## Scale & reliability notes (500+ students)

- Check-ins are de-duplicated using Google's fast cache (per session), so the system doesn't slow down as the sheet grows over the 10 days.
- The main load is the *rush* when a class starts. To smooth it, reveal the QR a couple of minutes into the session rather than before students settle — or leave it up for the first 10 minutes. The student page auto-retries on a transient error.
- If your University of Jordan account is **Google Workspace for Education** (very likely), execution quotas are comfortably higher than a personal Gmail account. Build it on your UJ account if you can.